‘the wave function evolves according to the Schrödinger equation.’ Some things don’t happen, because the Schrödinger equation never leads to them happening. For example, we will never see an electron spontaneously convert into a proton. That would change the amount of electric charge, and charge is strictly conserved. So branching will never create, for example, universes with more or less charge than we started with. Just because many things happen in Everettian quantum mechanics doesn’t mean that everything does.”

Alice’s father raised his eyebrows in skepticism. “Dear, you are surely nitpicking to save face. Maybe not strictly everything happens, but I believe it’s true that a great many crazy-sounding things do happen in various worlds, no?”

“Sure, I’m happy to admit that. Every time you run into a wall, the wave function branches into a number of worlds: some where you injure your nose, some where you harmlessly tunnel right through, and others where you bounce off and are thrown across the room, for example.”

“But that matters a lot, doesn’t it? In ordinary quantum mechanics the probability of a macroscopic object tunneling through a wall is not zero, but it’s unimaginably tiny, and we can just ignore it. In Many-Worlds, the probability is 100 percent that it happens in some world.”

Alice nodded, but her expression was that of someone who had gone over this ground many times before. “You’re absolutely right that this is a difference. But I would argue that it doesn’t matter a single bit. If you accept how Everettians derive the Born rule, you should *act as if* there is a probability of you tunneling through the wall, and that probability is so preposterously small that there’s no reason whatsoever to take it into consideration as you go through your everyday life. And if you don’t accept that argument, there is a much more serious worry about Many-Worlds for you to fret over.”

Her father was determined. “I think the issue of these low-probability worlds is important. What about those observers who, in the ensemble of Everettian worlds, end up seeing events that seemingly defy our Born rule predictions? If we measure a spin fifty times, there will be branches on which all of the results read spin-up, and others on which they all read spin-down. What are those poor observers supposed to conclude about quantum mechanics?”